

Boundary-Inversion Cosmology VI

A Sqrt-Reg Entropy Shield with a Stable EFT-Clock Completion

Technical Report and Numerical Stress-Test Package

Prepared for Lou | Current version: BIH-VI freeze

Core claim: BIH-VI contains a stable EFT-clock realization of a Sqrt-Reg entropy-selected cosmological background, with a Planck-compatible observable island and a controlled exit feature.

Status: technical framework freeze. Do not expand the model until the report has been reviewed.

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1. Executive summary

This report freezes the current state of the Boundary-Inversion Hypothesis at BIH-VI. Earlier branches were useful stress tests, but the final working branch is an effective-field-theory clock realization of the Sqrt-Reg entropy-selected background. The claim is deliberately narrow: BIH-VI is a stable EFT-level cosmological branch with falsifiable predictions, not a completed microscopic substrate theory.

1.1 Abstract

We develop a stable effective-field-theory realization of a Boundary-Inversion cosmological background. The geometric core is the invariant spherical marginal-horizon condition $\chi = h^{ab} \nabla_a R \nabla_b R = 0$, equivalently $R = 2G E_{MS}$, which unifies black-hole trapped and expanding cosmological anti-trapped sign sectors through opposite null-expansion orientation. Applying Hayward-style apparent-horizon thermodynamics motivates a Sqrt-Reg entropy functional whose equilibrium background caps both H^2 and the effective density. Several direct covariant completions fail as final microscopic actions: pure $f(T)$ suffers a scalar rank-changing constraint pathology, a minimal $f(T,B)$ repair fails to recover the correct scalar infrared limit, and an analytic $f(R)$ reconstruction gives the wrong sign for f_{RR} in the low-curvature branch. We then construct a stable EFT-clock completion realizing $H^2 = H_c^2 y(2-y)$, with scalar and tensor quadratic actions free of ghost and gradient instabilities. A cubic near-cap gear profile $m(y) = \mu(1-y)^3$ generates a Planck-compatible parameter island. For the benchmark $p = 3$, $\mu = 1$, $c_s = 1$, $N_* = 55$, and $z_e = 1/3$, the model predicts $n_s = 0.9652$, $r = 1.4e-3$, $\alpha_s = -5.8e-4$, and $H_c = 3.8e-6 M_{Pl}$. A Mukhanov-Sasaki stress test of the exit switch shows that $\Delta = 0.01$ keeps CMB pivot modes uncontaminated while producing a localized exit-scale feature. The result is a falsifiable EFT-level cosmological branch rather than a final microscopic substrate theory.

Table 1. Current BIH-VI decision table.

Component	Status	Reason
Geometric seam	Survives	$\chi = 0 / R = 2G E_{MS}$ is the invariant spherical-horizon seam.
Sqrt-Reg entropy	Survives	Caps H^2 and effective density in equilibrium horizon thermodynamics.
Pure $f(T)$	Rejected as final action	Background/tensors pass; scalar rank-changing constraint pathology.
Linear $f(T,B)$	Rejected as minimal repair	Preserves background; scalar IR behavior fails.
Analytic $f(R)$	Rejected	Reconstruction exists but f_{RR} has wrong sign in analytic IR branch.
BIH-VI EFT clock	Current best branch	Stable scalar/tensor quadratic action and viable CMB parameter island.
Exit switch	Passes first M-S stress test	Pivot-safe for Δ around 0.01; localized exit-scale feature.

2. Core equations and definitions

Natural units $c = \hbar = k_B = 1$ are used. The metric signature convention is $(-, +, +, +)$. All equations in this report use plain-text notation for portability across DOCX/PDF renderers.

2.1 Spherical geometry and null expansions

$$ds^2 = h_{ab} dx^a dx^b + R^2 d\Omega^2$$

$$\theta_{pm} = (2/R) l_{pm}^a \nabla_a R$$

normal: (+,-), trapped: (-,-), anti-trapped: (+,+)

2.2 Invariant seam

$$\chi = h^{ab} \nabla_a R \nabla_b R$$

$$\chi = 1 - 2G E_{MS}/R$$

$$\chi = 0 \Leftrightarrow R = 2G E_{MS}$$

$$\theta_+ + \theta_- = -4\chi/R^2 \quad \text{for } l_+ \cdot l_- = -2$$

2.3 Apparent horizon and entropy variables

$$x = H^2 + k/a^2$$

$$R_A = x^{-1/2}, \quad A = 4\pi / x$$

$$S_{\text{sqrt}}(A) = (1/(4G)) [\sqrt{A(A-A_c)} + A_c \text{arcosh} \sqrt{A/A_c}]$$

$$S_A = dS/dA = 1/[4G \sqrt{1-A_c/A}]$$

3. Boundary-Inversion seam

The map $B: (\theta_+, \theta_-) \rightarrow (-\theta_-, -\theta_+)$ is useful as orientation language, but it is not the invariant physical seam. The invariant seam is $\chi = 0$, equivalently $R = 2G E_{MS}$. Trapped and anti-trapped regions are distinguished by the sign sector of the future null expansions and the time orientation of the screen.

3.1 Schwarzschild and FLRW checks

$$\text{Schwarzschild: } \chi = 1 - 2GM/R$$

$$\text{FLRW: } \chi = 1 - (H^2 + k/a^2) R^2$$

$$\text{FLRW apparent horizon: } R_A = 1/\sqrt{H^2 + k/a^2}$$

This common $\chi=0$ condition is known spherical-horizon geometry, not a new theorem. The BIH contribution is to use it as the organizing seam connecting black-hole trapped screens and cosmological anti-trapped screens, then to construct a thermodynamic and EFT realization around the cosmological side.

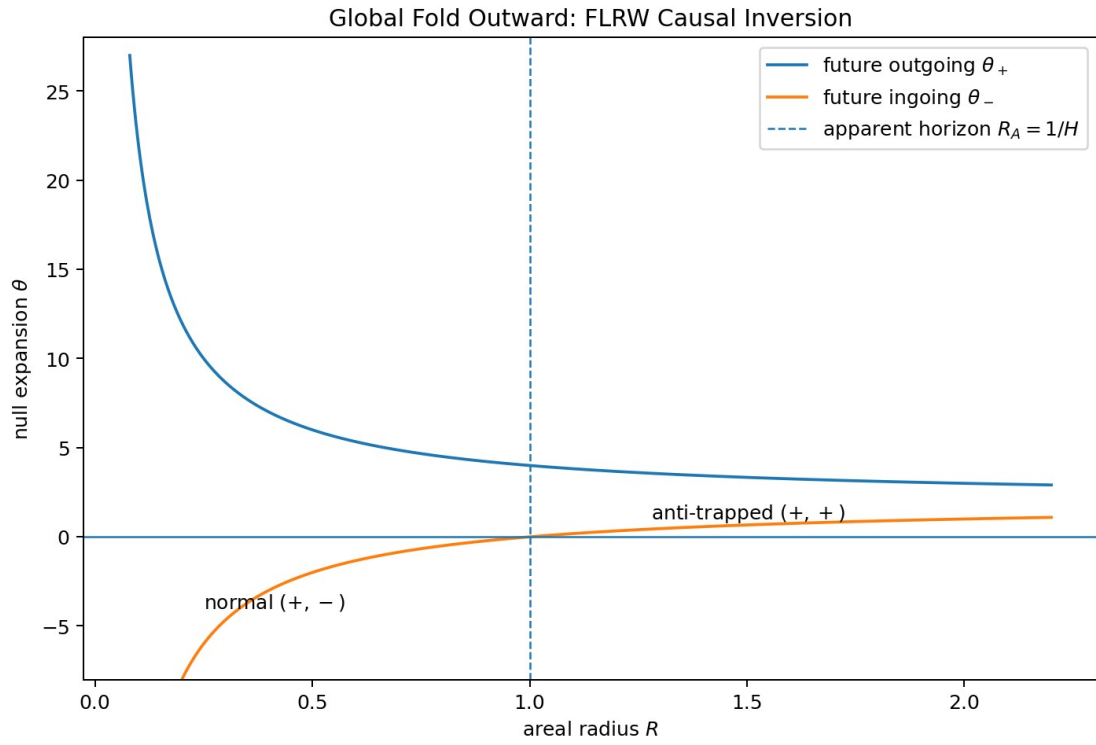


Figure 1. FLRW null-expansion sign sectors: normal to anti-trapped across the apparent horizon.

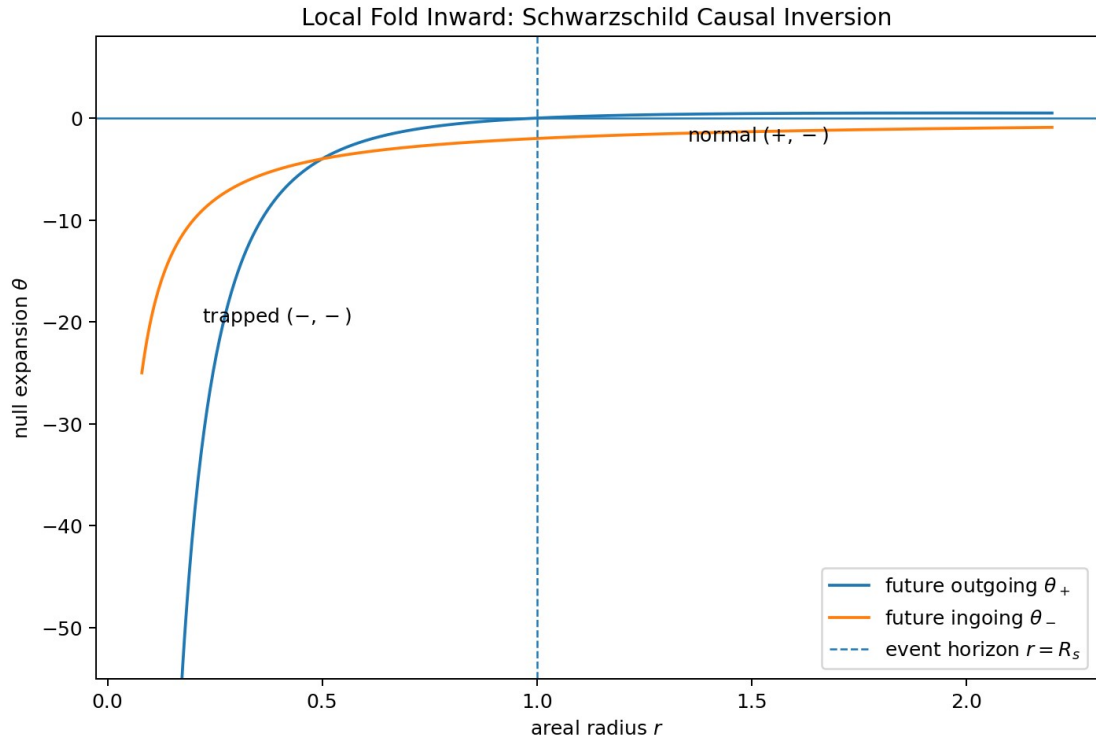


Figure 2. Schwarzschild null-expansion sign sectors: normal to trapped across the horizon.

4. Hayward thermodynamics and the Sqrt-Reg density shield

The thermodynamic layer uses the apparent horizon rather than a literal material boundary. In equilibrium, the Hayward-style entropy-production diagnostic reduces to a modified acceleration equation.

$$\Gamma = \dot{S}_i = 8 \pi^2 H R_A^4 (\rho + p) - 8 \pi R_A S_A \dot{R}_A$$

$$\Gamma = 0 \Rightarrow \dot{H} - k/a^2 = -\pi (\rho + p)/S_A$$

$$G_{\text{eff}}(A) = 1/(4 S_A) = G \sqrt{1 - A_c/A}$$

4.1 Sqrt-Reg Friedmann function

$$F_{\text{sqrt}}(x) = 2 x_c [1 - \sqrt{1 - x/x_c}]$$

$$F_{\text{sqrt}}(H^2) = 8 \pi G \rho / 3 \quad (\text{flat, } \Lambda = 0)$$

$$H^2 = (8 \pi G \rho / 3) [1 - 2 \pi G \rho / (3 H_c^2)]$$

$$\rho_{\text{max}} = 3 H_c^2 / (4 \pi G)$$

This is the key improvement over the earlier log-critical entropy branch. The log-critical branch capped H^2 but left the matter density divergent. The Sqrt-Reg branch caps both H^2 and the effective density.

5. Failed direct completions

This section is intentionally included. It documents the failure modes so the final branch is not presented as a lucky first guess.

5.1 Pure $f(T)$

Pure $f_{\text{sqrt}}(T)$ reconstructs the Sqrt-Reg background and passes the tensor-sector check, but its scalar/tetrad sector has a rank-changing antisymmetric constraint. The relevant coefficient is proportional to $f_{TT} \dot{T}$, which vanishes in the IR and is nonzero at finite u . That produces a discontinuous constraint structure and indicates strong-coupling pathology.

$$A_{\text{constraint}} = f_{TT} \dot{T}$$

$$A_{\text{constraint}} \rightarrow 0 \text{ as } u \rightarrow 0, \text{ but } A_{\text{constraint}} \neq 0 \text{ for generic } 0 < u < 1$$

5.2 Linear $f(T, B)$

A linear $Q(T)B$ repair can preserve the Sqrt-Reg background only if the T -only part is retuned. The minimal repair $f_q = g_{\text{sqrt}} + q (\tau/\tau_c)(B - 2\tau)$ preserves the background, but generically produces scalar-sector IR behavior analogous to an unscreened extra scalar and fails as a minimal final action.

$$f_q(\tau, B) = g_{\text{sqrt}}(\tau) + q (\tau/\tau_c)(B - 2\tau)$$

5.3 Analytic $f(R)$

Metric $f(R)$ reconstruction exists for the Sqrt-Reg background, but the analytic GR-limit branch gives the wrong sign for f_{RR} . For dust, the low-curvature expansion has an anti-Starobinsky sign.

$$f(R) = R - R^2/(54 H_c^2) + \dots \quad (\text{dust analytic branch})$$

$$f_{RR} < 0 \Rightarrow \text{scalaron instability}$$

6. BIH-VI stable EFT-clock completion

The stable branch realizes the Sqrt-Reg background inside the EFT of cosmological perturbations. This does not claim a final microscopic substrate; it gives a controlled action-level realization with healthy quadratic scalar and tensor sectors.

$$S_{\text{EFT}} = \int d^4x \sqrt{-g} \left[M_{\text{Pl}}^2 R/2 - \Lambda(t) - c(t) g^{\{00\}} + (M_2^4(t)/2)(\delta g^{\{00\}})^2 + \dots \right]$$

$$H^2 = H_c^2 y(2-y), \quad y_N = -m(y) y$$

$$\dot{H} = -m H_c^2 y(1-y)$$

$$\epsilon_H = m(1-y)/(2-y)$$

$$c(y) = M_{\text{Pl}}^2 m H_c^2 y(1-y)$$

$$\Lambda(y) = M_{\text{Pl}}^2 H_c^2 y[(6-m)+(m-3)y]$$

$$M_2^4 = c/2 (c_s^{-2} - 1)$$

6.1 Perturbation actions

$$S_{\text{zeta}}^{(2)} = M_{\text{Pl}}^2 \int dt d^3x a^3 (\epsilon_H/c_s^2) \left[\dot{z}^2 - c_s^2 (\text{grad } z)^2/a^2 \right]$$

$$S_T^{(2)} = (M_{\text{Pl}}^2/8) \int dt d^3x a^3 \left[\dot{h}_{ij}^2 - (\text{grad } h_{ij})^2/a^2 \right]$$

$$\text{stable if: } \epsilon_H > 0, c_s^2 > 0, c_T^2 = 1$$

7. Cubic gear observables and parameter island

$$z = 1-y$$

$$m(y) = \mu z^p$$

$$N_* = (1/\mu) \int_{z_*}^{z_e} dz/[z^p(1-z)]$$

$$\epsilon_* = \mu z_*^{p+1}/(1+z_*)$$

$$r = 16 \epsilon_* c_s$$

$$n_s - 1 = -2 \epsilon_* - \mu z_*^p (1-z_*)[(p+1)/z_* - 1/(1+z_*)]$$

The viable branch is centered near $p = 3$. This exponent gives the Starobinsky-like red-tilt scaling in the near-cap asymptotic limit, while finite exit effects shift the optimal μ toward μ about 1 for $N_* = 55$ and $z_e = 1/3$.

Table 2. BIH-VI benchmark line for $p = 3$, $N_ = 55$, $z_e = 1/3$, $c_s = 1$.*

μ	z_{star}	y_{star}	n_s	r	α_s	H_c/Mpl
0.5000	0.1372	0.8628	0.9682	0.0025	-4.737e-04	5.130e-06
0.7000	0.1176	0.8824	0.9665	0.0019	-5.312e-04	4.486e-06
1.0000	0.0991	0.9009	0.9652	0.0014	-5.784e-04	3.834e-06
1.5000	0.0811	0.9189	0.9643	9.623e-04	-6.165e-04	3.168e-06
2.0000	0.0702	0.9298	0.9638	7.274e-04	-6.351e-04	2.752e-06

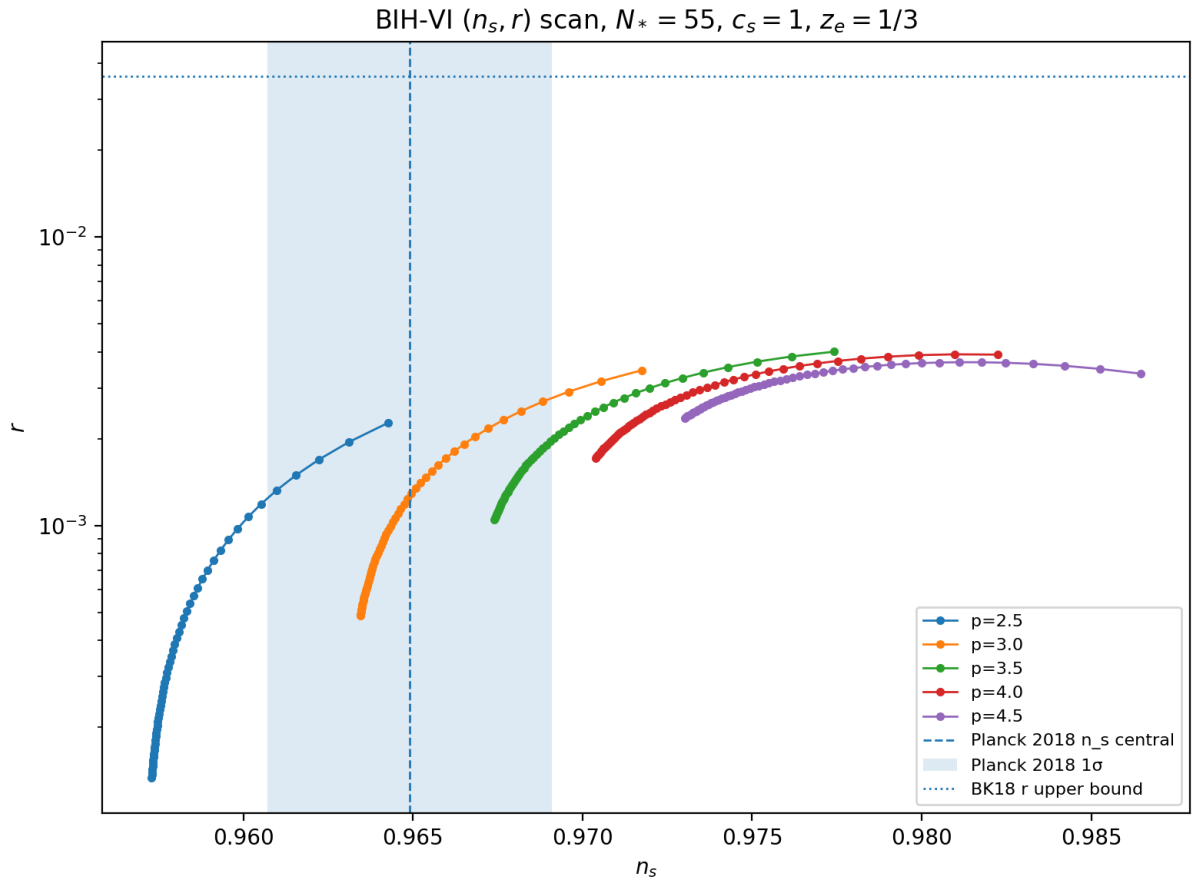


Figure 3. BIH-VI (n_s, r) parameter scan for $N_* = 55, c_s = 1, z_e = 1/3$.

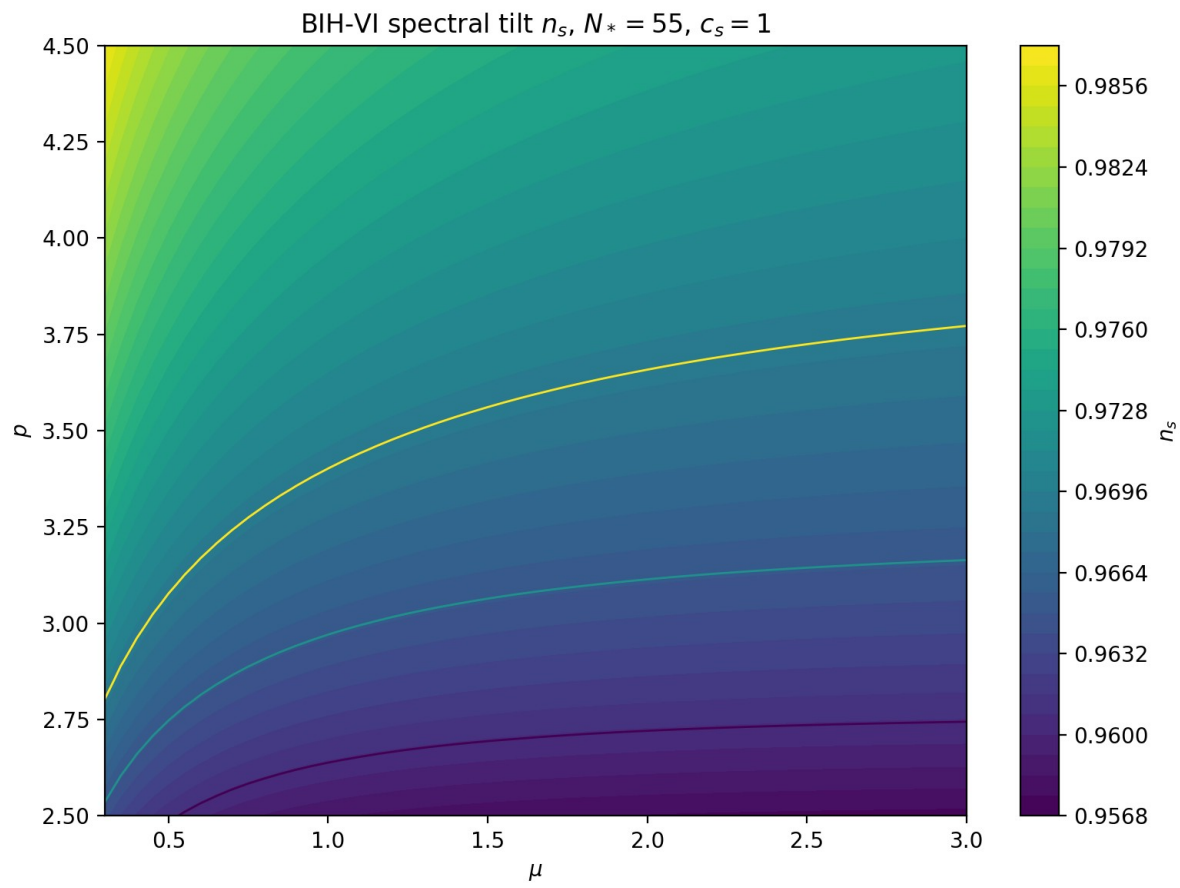


Figure 4. Spectral tilt n_s over the (μ, p) plane.

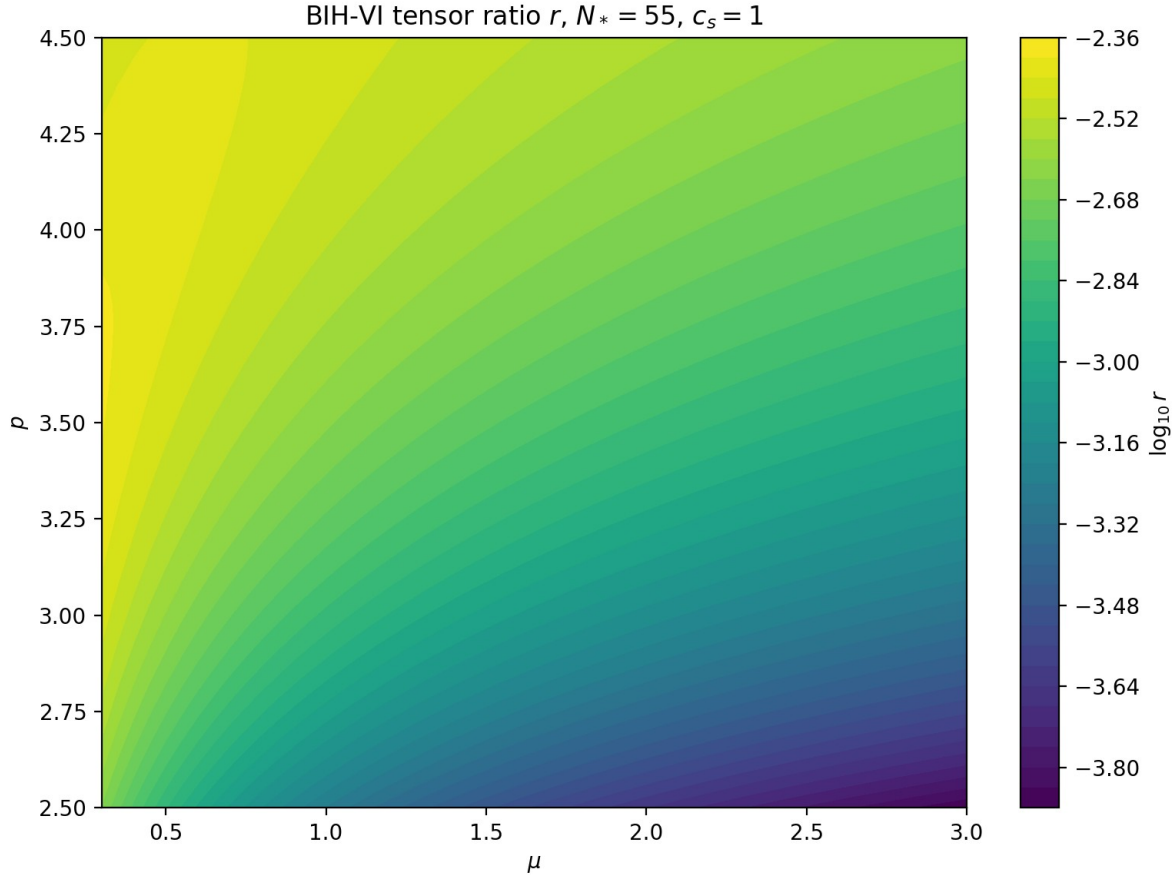


Figure 5. Tensor-to-scalar ratio r over the (μ, p) plane.

8. Exit switch and Mukhanov-Sasaki stress test

$$m(y) = \mu(1-y)^3 + m_{\text{post}} S(y)$$

$$S(y) = 1/[1 + \exp((y-y_g)/\Delta)]$$

$$\text{benchmark: } \mu=1, p=3, y_g=2/3, \Delta=0.01, m_{\text{post}}=4$$

The switch is designed to be invisible at the CMB pivot while forcing a handoff to a radiation-like phase near the end of inflation. The slow-roll estimate predicts a localized feature near the switch; the Mukhanov-Sasaki calculation checks whether that feature is physical and numerically stable.

$$v_k'' + [c_s^2 k^2 - z_s''/z_s] v_k = 0$$

$$z_s = a M_{\text{Pl}} \sqrt{2 \epsilon_H}/c_s$$

$$P_\zeta(k) = (k^3/(2 \pi^2)) |v_k/z_s|^2$$

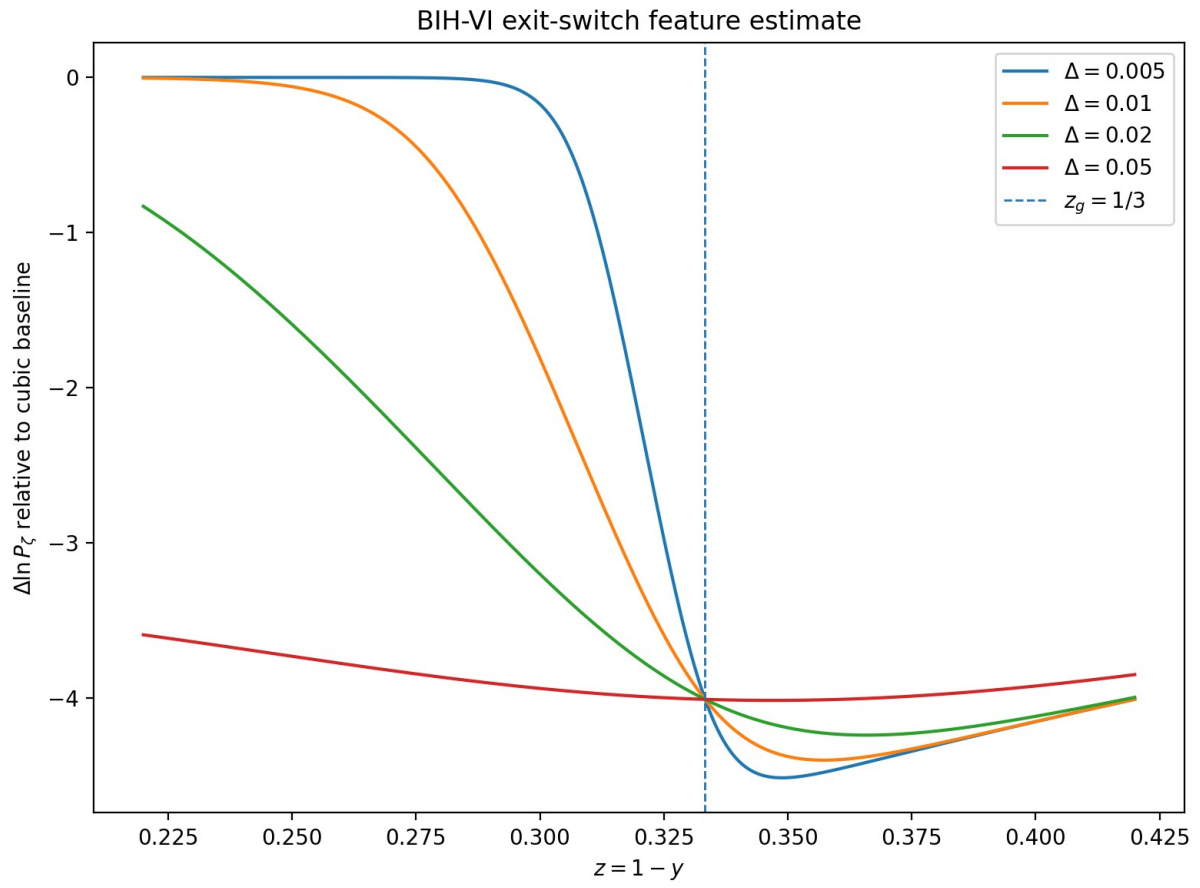


Figure 6. Slow-roll exit-feature estimate: switch-induced $\Delta \ln P_z$ relative to the cubic baseline.

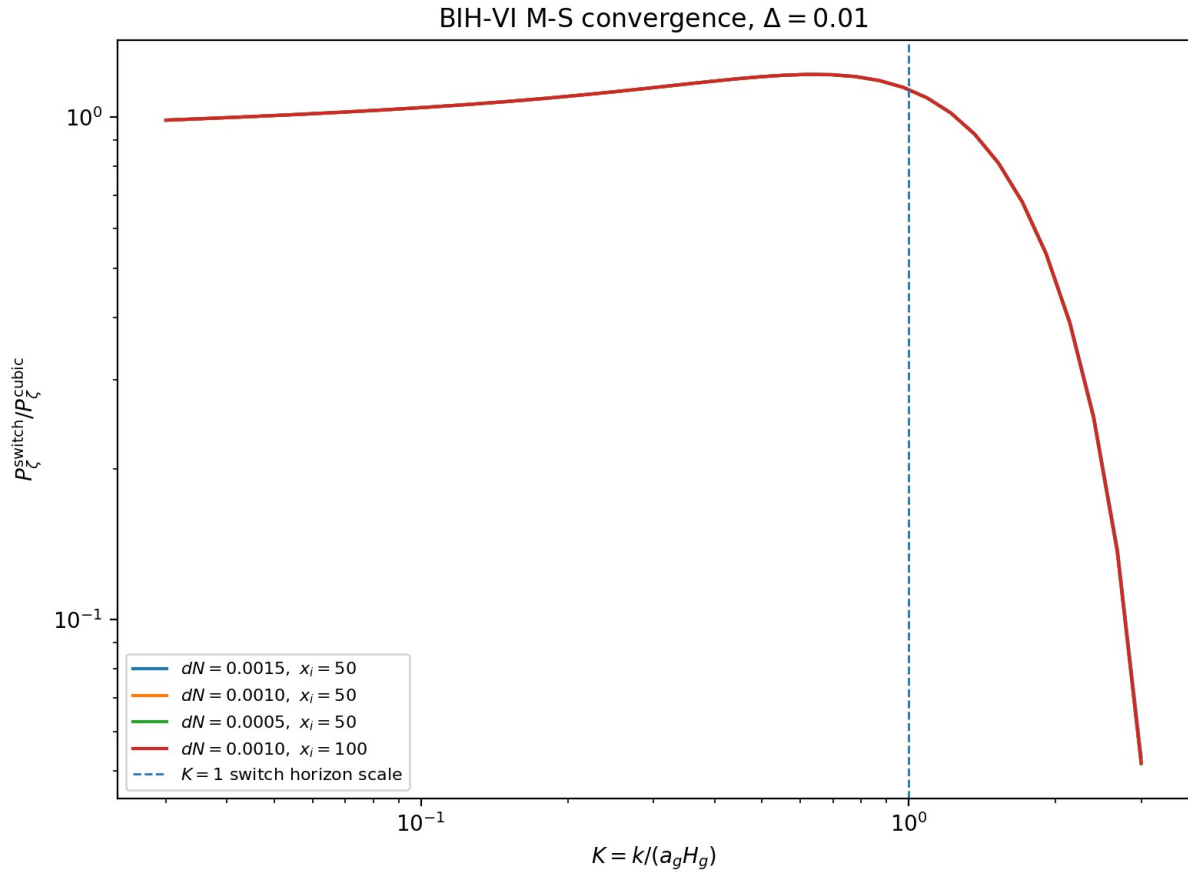


Figure 7. Mukhanov-Sasaki switch-feature convergence for $\Delta = 0.01$.

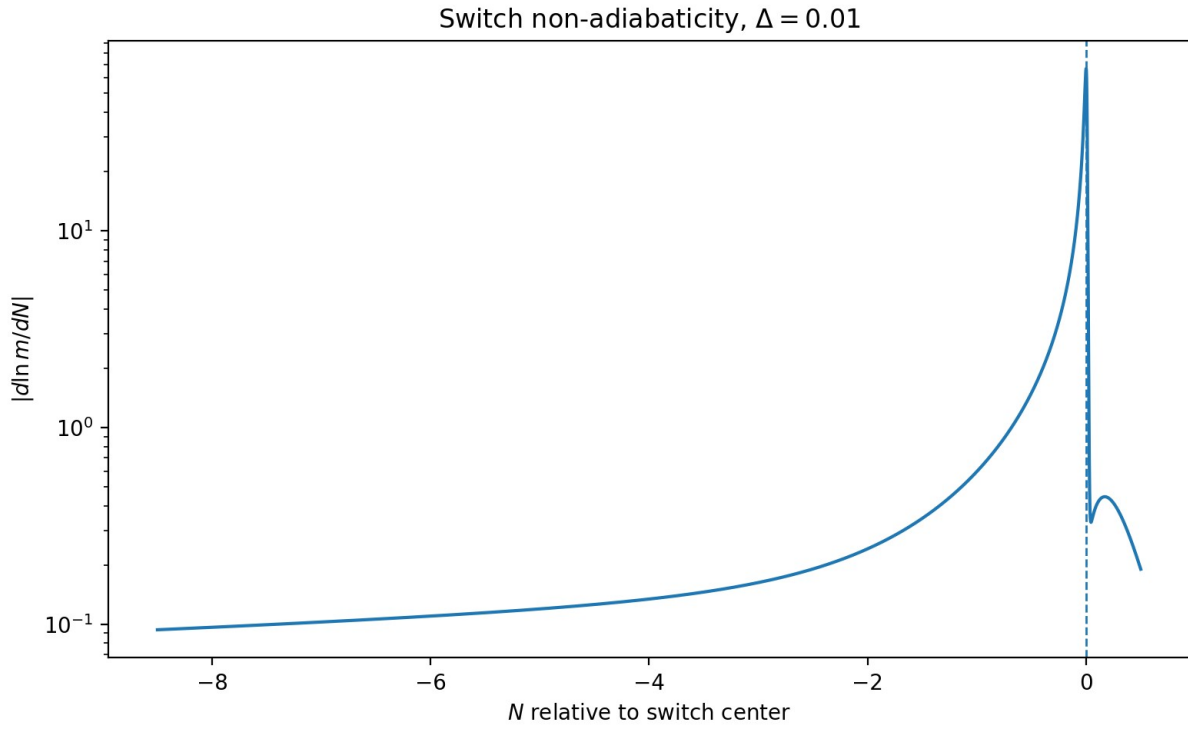


Figure 8. Switch non-adiabaticity $|d \ln m/dN|$ for $\Delta = 0.01$.

Table 3. M-S convergence summary for Delta = 0.01.

config	min_ratio_all	max_ratio_all	min_ratio_K_l t_1	max_ratio_K_l t_1	min_ratio_K_g t_1	max_ratio_K_g t_1
\$dN=0.0015,\backslash\$ x_i=50\$	0.0532	1.2167	0.9865	1.2167	0.0532	1.0934
\$dN=0.0010,\backslash\$ x_i=50\$	0.0519	1.2165	0.9866	1.2165	0.0519	1.0924
\$dN=0.0005,\backslash\$ x_i=50\$	0.0517	1.2164	0.9865	1.2164	0.0517	1.0921
\$dN=0.0010,\backslash\$ x_i=100\$	0.0519	1.2166	0.9865	1.2166	0.0519	1.0927

9. Engineering interpretation: laminar flow and controlled exit feature

The fluid language is useful as an engineering guide, not as an additional postulate. The formal model does not require a literal oil, material exterior, or biological interpretation. The analogy maps as follows:

Table 4. Engineering analogy translated into technical BIH-VI objects.

Intuition	Technical object
black oil / structured medium	EFT clock sector
surface tension	Sqrt-Reg cap H_c
viscosity	c_s, M_2^4, and higher-order EFT operators
laminar flow	stable evolution of CMB-pivot modes
turbulence / splash	localized exit feature in P_zeta(k)
wind tunnel test	Mukhanov-Sasaki convergence check

The disciplined version is: BIH-VI is laminar through the CMB window and locally turbulent only at the exit boundary. The turbulence is a controlled localized feature rather than a catastrophic instability in the tested parameter range.

10. Falsifiability and forecast targets

BIH-VI is falsifiable through its observable island and its exit feature. The benchmark prediction is small but not arbitrarily small: r is around 10^{-3} . LiteBIRD-level sensitivity is therefore the natural future comparison.

$$\text{fiducial: } n_s = 0.9652, \quad r = 1.4e-3, \quad \alpha_s = -5.8e-4$$

localized exit feature: $P_{\text{switch}}/P_{\text{cubic}} \sim 1$ for $K \ll 1$, modest ringing near $K \sim 1$, suppression for high K

Table 5. Falsifiability routes.

Test	Falsification route
Tensor ratio	Future constraints push well below $r \sim 10^{-3}$ without detection.
Spectral tilt	Preferred n_s moves outside the BIH-VI island after nuisance/reheating freedom is accounted for.
Running	A large or opposite-sign α_s is detected.
Exit features	Small-scale data rule out the predicted localized suppression/ringing region.
Microscopic completion	No covariant stable completion can reproduce the EFT branch without hidden instabilities.

11. Limitations and scope boundaries

This section defines the scope boundaries of BIH-VI. They are not treated as failures; they are explicit validity limits of the present technical note. The report establishes a stable EFT-level realization and a phenomenological observable island. It does not claim a unique microscopic substrate theory.

11.1 EFT vs. microscopic completion gap

Shortfall. BIH-VI currently provides a robust effective-field-theory completion of the Sqrt-Reg background, not a derivation from a unique UV-complete microscopic Lagrangian. The Sqrt-Reg entropy functional and the EFT clock demonstrate that a stable branch exists, but they do not identify the fundamental degrees of freedom that generate the entropy law.

Report stance. The model is therefore framed as a phenomenological EFT branch: validity-limited, stable at the level studied, and suitable for observational testing. It should not be advertised as the only possible UV completion of the Boundary-Inversion/Sqrt-Reg boundary.

11.2 Sensitivity to switch dynamics

Shortfall. The transition profile $S(y)$ is parameterized rather than derived. The benchmark $\Delta = 0.01$ is pivot-safe and numerically stable in the first Mukhanov-Sasaki stress test, but the switch width, location, and post-inflationary handoff are not yet forced by the Sqrt-Reg entropy law.

Report stance. The exit feature should be described as a testable signature of transition-width selection. The switch is an open parametric sensitivity, not a hard-coded fundamental constant. A future microscopic completion should explain why a particular Δ , y_g , and m_{post} are selected.

11.3 Single-clock limitation

Shortfall. BIH-VI is a single-clock EFT construction. The real early universe may involve multiple fields, entropy/isocurvature perturbations, reheating-sector couplings, or dark-sector interactions that change the scalar spectrum or the exit feature.

Report stance. The present model is a minimalist stability construction. It proves that a pivot-safe primary adiabatic scalar spectrum can be obtained from the Sqrt-Reg background, but it does not claim to explain all primordial anomalies or all possible multi-field signatures.

11.4 Numerical resolution of the exit feature

Shortfall. The Mukhanov-Sasaki switch calculation is a direct power-spectrum proxy. It shows convergence under the tested time steps and initial depths, but it is not a full Boltzmann transfer calculation into final CMB temperature and polarization spectra.

Report stance. The M-S feature is a controlled first stress test. Mapping it to observed angular spectra, matter power, 21-cm signatures, or induced gravitational-wave backgrounds requires full transfer functions and reheating physics.

11.5 Additional open work

- Non-Gaussianity and loop/backreaction constraints remain to be computed.

- Spatial curvature, anisotropy, and black-hole-sector implications remain outside the BIH-VI freeze.
- The exact source-coupled interpretation of the Sqrt-Reg density law remains distinct from the stable Einstein-frame EFT-clock realization.
- A future UV completion should derive the entropy functional, the clock profile $m(y)$, and the exit switch from deeper microphysics rather than fitting them phenomenologically.

11.6 Scope-boundary statement

This work focuses on the EFT-level stability and observational viability of the BIH-VI branch. It does not attempt to solve the UV completion of the Hayward/Sqrt-Reg entropy functional, nor does it model the full multi-field transfer functions. Its central result is the existence of a stable, boundary-anchored cosmological EFT with a Planck-compatible parameter island and a controlled exit feature.

Appendix A. Equation map

Geometric seam:

$$\chi = h^{\{ab\}} \nabla_a R \nabla_b R$$

$$\chi = 1 - 2G E_{MS}/R$$

$$\chi = 0 \Leftrightarrow R = 2G E_{MS}$$

Null expansion sectors:

$$\text{normal: } (+, -)$$

$$\text{trapped: } (-, -)$$

$$\text{anti-trapped: } (+, +)$$

Hayward equilibrium:

$$\Gamma = 0$$

$$\dot{H} - k/a^2 = -\pi(\rho + p)/S_A$$

Sqrt-Reg entropy:

$$S_A = 1/[4G \sqrt{1 - A_c/A}]$$

Sqrt-Reg Friedmann:

$$F(x) = 2x_c [1 - \sqrt{1 - x/x_c}]$$

$$F(H^2) = 8\pi G \rho/3$$

Density shield:

$$H^2 = (8\pi G/3)\rho [1 - 2\pi G \rho/(3H_c^2)]$$

$$\rho_{\max} = 3H_c^2/(4\pi G)$$

EFT-clock:

$$H^2 = H_c^2 y(2-y)$$

$$y_N = -m(y)y$$

$$\epsilon_H = m(1-y)/(2-y)$$

EFT functions:

$$c(y) = M_{Pl}^2 m H_c^2 y(1-y)$$

$$\Lambda(y) = M_{Pl}^2 H_c^2 y[(6-m) + (m-3)y]$$

$$M_2^4 = c/2(c_s^2 - 1)$$

Scalar action:

$$S_{\text{zeta}}^2 = M_{\text{Pl}}^2 \int a^3 \epsilon_H / c_s^2 [z_{\text{dot}}^2 - c_s^2 (\text{grad } z)^2 / a^2]$$

Gear:

$$m(y) = \mu(1-y)^p$$

Benchmark:

$$p=3, \mu=1, c_s=1, N_*=55, z_e=1/3$$

Predictions:

$$n_s=0.9652$$

$$r=1.4e-3$$

$$\alpha_s=-5.8e-4$$

$$H_c=3.8e-6 M_{\text{Pl}}$$

Exit:

$$S(y) = 1/[1 + \exp((y-y_g)/\Delta)]$$

$$y_g=2/3, \Delta=0.01$$

Mukhanov-Sasaki:

$$v_k'' + [c_s^2 k^2 - z_s''/z_s]v_k=0$$

Appendix B. Numerical artifact index

Table B1. Numerical artifacts included in the package.

File	Description
bih_vi_ns_r_scan_clean.png	Parameter scan in the (n_s,r) plane.
bih_vi_ns_heatmap_clean.png	n_s heatmap over (mu,p).
bih_vi_r_heatmap_clean.png	r heatmap over (mu,p).
bih_vi_parameter_scan_clean.csv	Full parameter scan table.
bih_vi_benchmark_table.csv	Benchmark values for p=3.
bih_vi_scan_summary.csv	Viable-region summary.
bih_vi_exit_feature_dlnP.png	Slow-roll exit-feature estimate.
bih_vi_exit_switch_stress_test.csv	Switch width stress test table.
bih_vi_ms_convergence_delta001.png	M-S convergence plot for Delta=0.01.
bih_vi_ms_convergence_summary.csv	M-S convergence summary.
bih_vi_ms_convergence_diffs.csv	M-S convergence differences.

Appendix C. References

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End of BIH-VI technical report freeze.